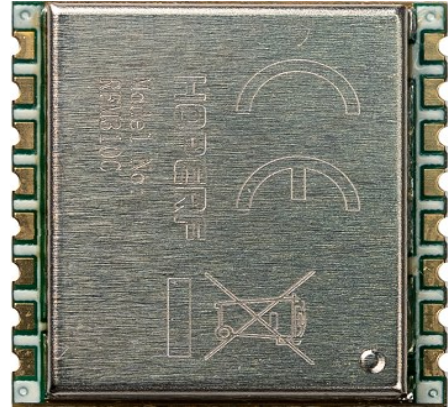


RFM310CW Wireless Transceiver Module

Product Overview

The RFM310CW module is a low-power, high-performance RF transceiver module for wireless applications that supports OOK, (G)FSK, and 4(G)FSK modulation. It supports multiple packet formats and encoding/decoding methods to flexibly meet a variety of application requirements. Features including abundant GPIO and interrupt configurations, Duty-Cycle operation mode, channel sensing, high-precision RSSI, low voltage detection, power-on reset, low-frequency clock output, fast frequency hopping, and squelch output enable more flexible applications.



Product Information

- Super anti-interference ability, suitable for use in complex interference environments
- Sensitivity-114dBm, DR = 10kbps, DEV = 5KHz @433.92 MHz
- Operating frequency: 868MHz
- Power supply voltage input range: 1.8 V-3. 6V
- Output power: +13dbm @868MHz
- Transmitting working current: 32 mA @13dbm
- Received operating current: 10.4mA (DCDC enabled)
- Automatic frequency correction (AFC)
- Fast and accurate effective signal monitoring (PJD, RSSI)
- Automatic ACK and retransmission
- 4-wire SPI interface
- Supports pass-through and packet modes

Application

- Intelligent metering
- Home security and building automation
- Industrial monitoring and control
- Wireless sensor node
- Tag reader/writer
- ISM band data communication

1. Module Pin Arrangement

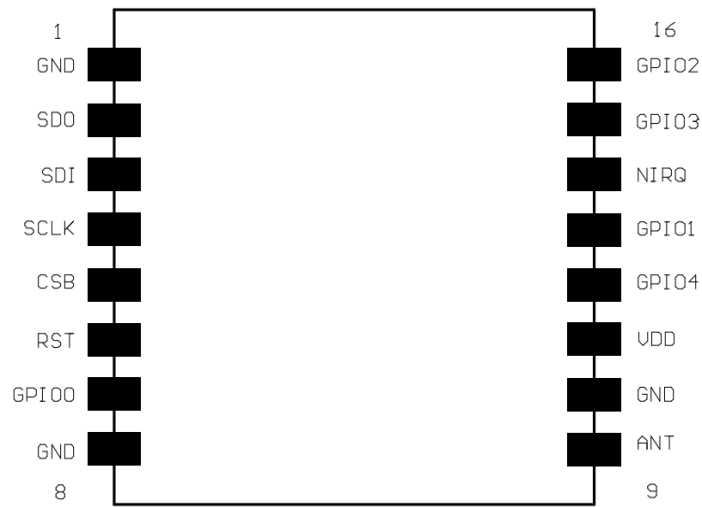


Figure 1. Front View of RFM310CW Module

2. Module Pin Definition

Table 1. RFM310CW Module Pin Definition

Pin	Name	Function description
1	GND	Ground
2	SDO	SPI Data Output
3	SDI	SPI Data Input
4	SCLK	SPI data clock
5	CSB	SPI chip selection input
6	GPIO5/RST	IO, configurable
7	GPIO0	IO, configurable
8	GND	Ground
9	ANT	Antenna port
10	GND	Ground
11,	VDD	Power supply
12	GPIO4	IO, configurable
13	GPIO1	IO, configurable
14	NIRQ	IO, configurable
15	GPIO3	IO, configurable
16	GPIO2	IO, configurable

3. Module Electrical Parameters

Test conditions: Power supply 3.3V, working temperature 25°C.

Table 2. Module Electrical Parameters

Parameter	Symbol	Status	Minimum	Typical	Maximum	Unit
Operating frequency	F _c	RFM310CW-868S2		868		MHz
Receive sensitivity	S	FSK: DR = 10kbps, DEV = 5KHz		-109		dBm
Operating voltage	V _{DD}		1.8	3.3	3.6	V
Receive operating current	I _{Rx}	Enable DCDC		10.4		mA
Transmit power	P _{out}	VCC=3.3V		13		dBm
Transmitting working current	I _{Tx}	Enable DCDC 13dbm		32		mA
Sleep current	I _{Sleep}	Duty Cycle = OFF		0.6		uA
Operating temperature	T _{OP}		-40		+85	°C

Note:

The operating frequency of the module needs to be configured and corrected via the RFPDKF software. The default value of the Xtal Cap Load option is 2, and this value needs to be changed to 31 when using the 868MHz frequency.

4. Module Outline Dimension

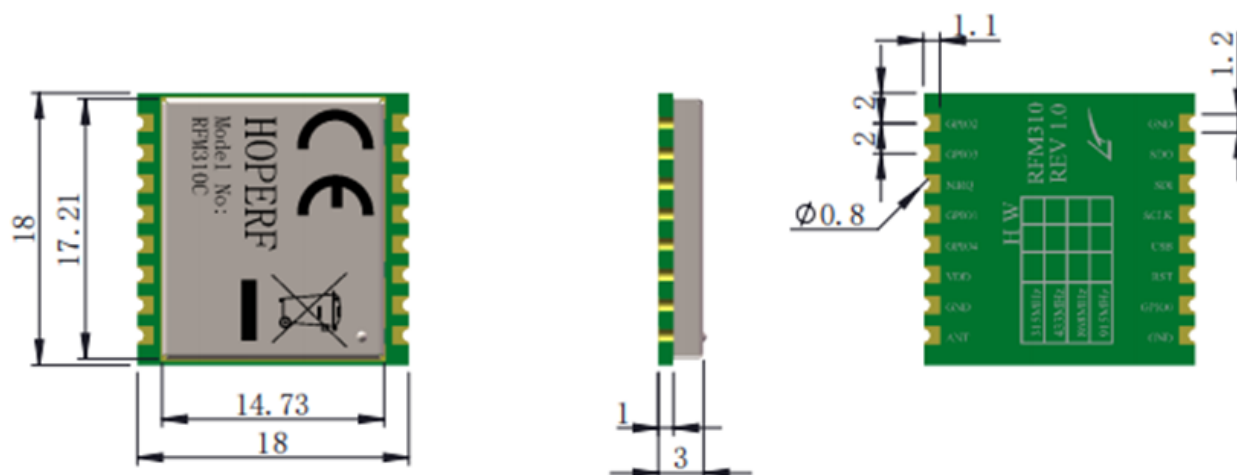


Figure 2. Module Outline Dimension

5. Reflow Temperature Profile

We recommend that the reflow soldering temperature profile be set in compliance with relevant IPC standards and configured according to the actual profile of the solder paste.

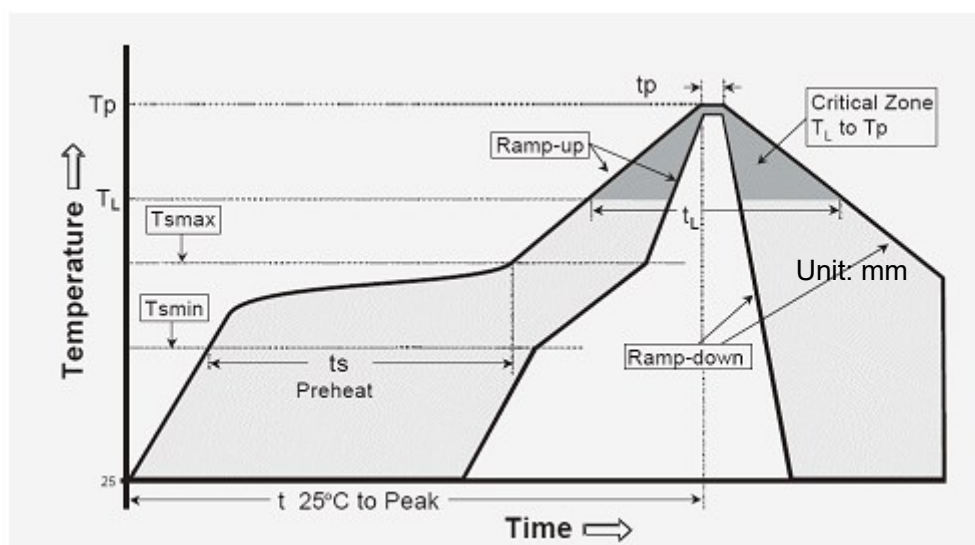


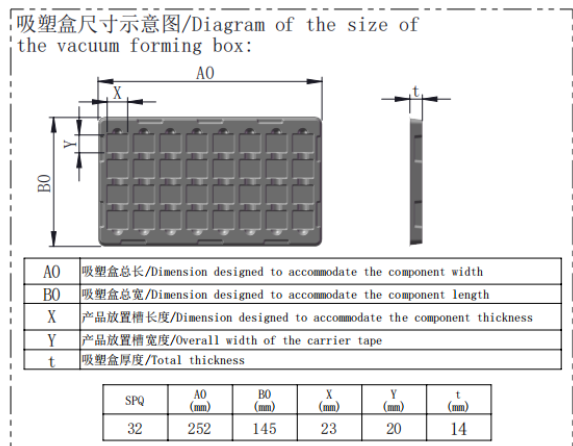
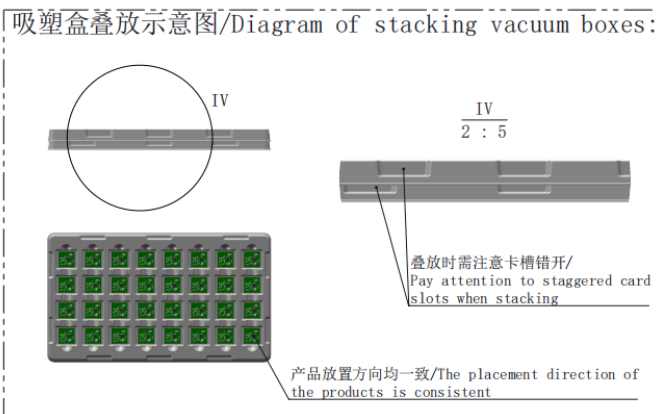
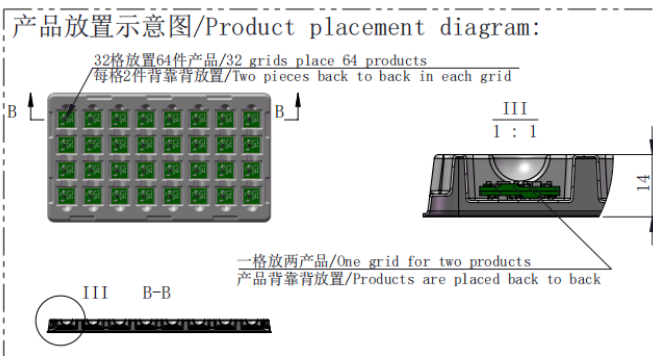
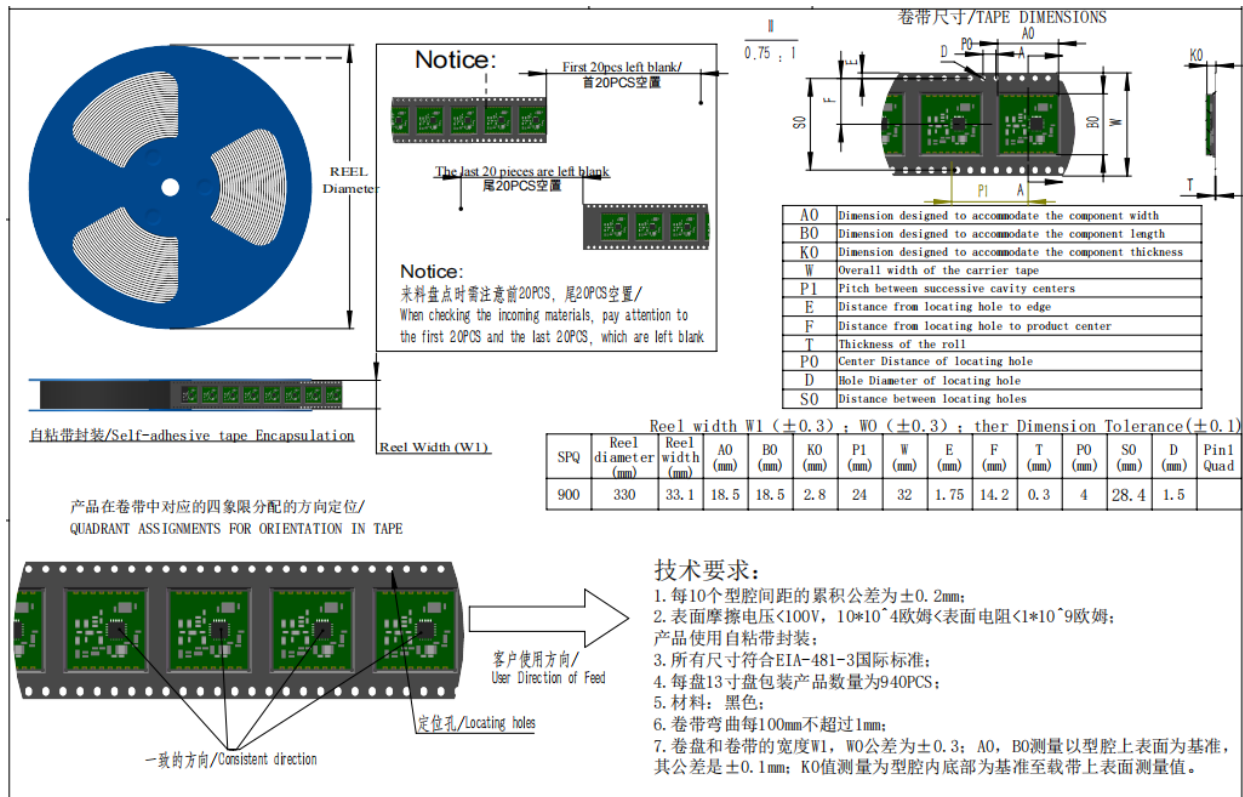
Table 3. Temperature and Time Parameters

Preheating temperature – Lower Limit Temperature (T _{smin}) – Upper Limit Temperature (T _{smax}) – Warm-up time (ts)	150°C 200°C 60~180 seconds
Heating rate from preheat to peak temperature (TL-Tp)	3°C/sec (max)
Welding time – Welding temperature – Time above melting point	60~90 seconds 217°C 60~150 seconds
Peak temperature	240+/-5°C
Welding time within peak temperature $\pm$ °C range	10~30 seconds
Cooling rate	6°C/sec (max)
Full welding time	8 minutes (max)

Notes:

- Lead-free soldering materials have poorer wettability than tin-lead eutectic materials and require a higher melting zone temperature. To achieve better soldering quality and protect solder joints from oxidation, nitrogen protection is required during reflow soldering.
- For double-sided SMT products, avoid passing the module through reflow soldering with its component side facing down. The module shall be soldered in the second reflow process to prevent the components on the module from falling off or shifting.
- The module requires full and sufficient solder coverage. It is recommended to use a stencil thickness of 0.15 mm or above for the stepped stencil.
- When module rework soldering is necessary, manual soldering is recommended to avoid potential damage to the module caused by repeated reflow soldering processes.
- Customers shall formulate a soldering process compliant with the above guidelines according to actual production conditions.

6. Packaging Information: Reel tape, Blister box



技术要求:

1. 脱模斜度: 5° ;
2. 未注公差: $\pm 3\text{mm}$;
3. 所标注的高度尺寸均为上表面之间值 (除注明外);
4. 所标注的长宽尺寸均为根部尺寸 (除注明外);
5. 加标注识;
6. 未注明圆角为R3;
7. 材质: PS黑色, 厚度为0.8mm。

Technical requirements:

1. Taper for demolding: 5° ;
2. Unmarked tolerance: $\pm 3\text{mm}$;
3. The height dimensions marked are all values between the upper surfaces (Except as noted);
4. The length and width dimensions marked are all root dimensions (Except as noted);
5. Add annotation;
6. Undocumented rounding is R3;
7. Material: PS Color: Black, Thickness is 0.8mm.

7. Ordering Information

Table 4. Order Model

Module Model	Operating Frequency
RFM310CW-868S2 (R)	868MHz

8. Document Change Record

Table 5. Document Change Record

Version	Change Description	Date
V1.0	Initial	2026.8.26
V1.1	Text formatting adjustment	2026.9.24

9. Contact Information

Shenzhen Hope Microelectronics Co., Ltd

Address: 30th Floor, Building 8A, Phase III, Vanke Cloud, Xili Sub-district, Nanshan District, Shenzhen

Telephone: +86-0755-82973805

Fax: +86-0755-82973550

Email: sales@hoperf.com

Website: <http://www.hoperf.com>